

PAPER_692: M51 Whirlpool Galaxy: Tidal Arm Formation and UQFF Star Formation

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Class: M51WhirlpoolTidalInteraction

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Abstract

The Whirlpool Galaxy (M51a / NGC 5194) and its companion M51b (NGC 5195) form a spectacular interacting pair 23 Mly away. Tidal forces from M51b drive the prominent spiral arm structure and enhance star formation rates.

1. System Parameters

Parameter	Value
M_{M51a}	$1.6 \times 10^{11} M_{\odot}$
M_{M51b}	$2 \times 10^{10} M_{\odot}$
Separation	7.5 kpc
SFR	$3 M_{\odot}/\text{yr}$

2. Tidal Force

$$F_{tidal} = \frac{2GM_{comp}R_d}{d_{sep}^3}$$

This differential force stretches M51a's disk and drives arm formation.

3. Spiral Arm Pitch Angle

$$i_{pitch} = \arctan \left(\frac{F_{tidal}}{v_c^2/R} \right)$$

4. UQFF MUGE for M51

$$g_{M51} = \frac{G(M_a + M_b)}{r^2} (1 + H(z)) + \rho_{ISM} V g \frac{\rho_{SCm}}{\rho_{UA}}$$

5. UQFF Star Formation Efficiency

$$\text{SFE}_{UQFF} = \text{SFR} \left(1 + \frac{F_{tidal}}{g_{grav}} \right) (1 + f_{TRZ})$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Salo & Laurikainen (2000), MNRAS, 319, 393
- UQFF Framework, Star-Magic Session 174

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